Cisco Discovery Protocols CDP

Cisco discovery Protocol (CDP) is a layer 2 network protocol that lets cisco devices share basic information with each other. Informations such as device model, IP address, interface details, and software version. CDP runs automatically on cisco routers, switches, and other equipment, helping network engineers quickly discover what devices are connected and how they are linked, even if the network is large or poorly documented.

Operation: CDP works by having cisco device periodically send out small information packets (called CDP advertisements) from its active interfaces. These packets contain details such as the device's name, model, IP address, and the port its using. Any directly connected cisco device that receives these packets stores the information in a local CDP table.

When you run commands like "show cdp neighbors", the device displays the information it has collected from those advertisements. This makes CDP operate like a local listening and reporting system: devices broadcast their identity, neighbors listen, and you can view the stored data to understand how everything is connected. CDP does not pass information through multiple hops, it only work with directly connected neighbors making it reliable for mapping physical links and troubleshooting wiring issues.